

**S226 PRT661 DATA SCIENCE PRACTICE****Assessment 3 Group Project Presentaion Individual Contribution
Danala Group 8 Theme 2**

Australian NEM Price and Demand Analysis and Forecasting System (Docker, Airflow, PostgreSQL, Streamlit)

Group Members

1. DIPESH SEDHAI	S395457	S395457@students.cdu.edu.au
2. DILLIRAM DHAKAL	S396633	S396633@students.cdu.edu.au
3.NITESH RAJ CHAUDHARY*	S394811	S394811@students.cdu.edu.au
4.ROSHAN KUMAR SHRESTHA	S395498	S395498@students.cdu.edu.au

GitHub repository:

<https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>

Jira project board:

<https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list>

Youtube Link:

<https://www.youtube.com/watch?v=DIrxi1R78ys>

October 2026

Student: Nitesh Raj Chaudhary Student ID: S394811 Group: Danala Group 8 (Theme 2)

Project: Australian NEM Price and Demand Analysis and Forecasting System (Docker, Airflow, PostgreSQL, Streamlit)

1. Responsibilities

- Technical lead for data engineering, Docker/Postgres warehouse and pipeline design overall.
- Ownership of extraction and cleaning logic, integration of transformation and modelling in the pipelines.
- Architecture direction: Operational/Daily (:8501), Strategic Insights (:8502).
- Github repository owner/owner of main collaboration project and reproducible setup documentation (README / SETUP).
- The implementation of two ETL paths (nem_strategic_history, nem_operational_daily) and Airflow DAGs.

2. Work completed since Assessment 2

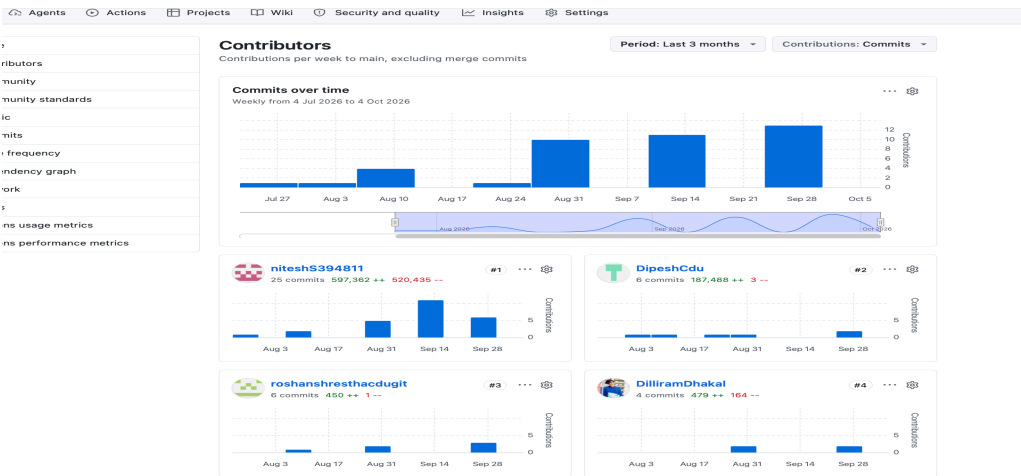
- Scaffold, Docker Compose, dependencies, postgres oriented setup (scaffold project, docker compose setup, postgres schemas, updated docker compose stack, dependencies).
- The fields for extended price, demand, interchange and generation data were added to NEMOSIS extract/clean.
- Operational daily incremental pipeline scripts aligned with the ops + analytics logic (strategic history).
- Strategic history and daily load (daily analytics load wiring) are supported by Airflow DAGs.
- Daily incremental load and handling dates when extracting.
- Repo cleanup/refactor (code cleanup, refactor code base and clean, setup guide updates).updates).

3. Evidence links

- GitHub: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>
- Jira : <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list?jql=project+=KAN+ORDER+BY+cf%5B10019%5D+ASC&atOrigin=eyJpIjoiYmlzMTBjN2JmZTQ2NGU2N2I0MGQ1MjUzZGMzMmZmYjUuLCJwIjoiaj9>
- Supporting artefacts: Assessment 3 PPT; <https://www.youtube.com/watch?v=Dirxi1R78ys>

4. Planned work before Assessment 4

- Stabilise full strategic load path.
- Refine both dashboards, add uniform demo narrative and screenshots for final report.
- Strengthen evaluation write-up (walk-forward, demand vs price stories).
- Final README one-command demo and contribution evidence pack for Assessment 4.



Page 2 of 2 — Lab evidence and reflection

Student: Nitesh Raj Chaudhary Student ID: S394811

Lab 2 (Week 4) — Module 8: Storing and Organizing Data

Lab title: Storing and Analyzing Data by Using Amazon Redshift

176492 > Modules > Module 8 Storing and Organizing Data > Lab: Storing and Analyzing Data by Using Amazon Redshift

Lab overview and objectives

Of the members of the data science team, Mary most often works with large datasets and columnar data stores. You have decided to explore using Amazon Redshift for your next proof of concept (POC) to address these use cases for her.

You ask Mary if she has a sample dataset that you can use for your POC. She suggests that you use a large music ticket sales dataset that she has worked with before. The dataset is available from a public Amazon Simple Storage Service (Amazon S3) bucket.

Data analysts often work with large datasets. For example, some data warehouses can contain multiple petabytes of data, and data lakes can be even larger. Amazon Redshift is designed to handle large datasets. Unlike traditional relational database management systems, Amazon Redshift uses columnar data storage. Columnar storage improves query performance and

Total score 30/30

TASK 2 - Cluster created 5/5

TASK 2b - Security group associated 5/5

TASK 3 - Tables created and data loaded 5/5

TASK 4 - Table data queried 5/5

TASK 5 - Queried by using the AWS CLI 5/5

TASK 7 - IAM user queried data 5/5

Submission Details

Last submitted: Sep-29-2026 4:49:12 am PDT

Submission count: 1

Due date: None

Lab: Storing and Analyzing Data by Using Amazon Redshift

Grades for s394811@students.cdu.edu.au

Arrange By

Due Date

Apply

Name	Due	Submitted	Status	Score
Course assessment Assignments				0 / 100
Module 2 Knowledge Check Assignments		Aug 20 at 1:21pm		0 / 100
Module 3 Knowledge Check Assignments		Aug 20 at 1:27pm		0 / 100
Module 4 Knowledge Check Assignments		Sep 13 at 2:44pm		0 / 100
Module 5 Knowledge Check Assignments		Sep 29 at 11:33am		0 / 100
Module 6 Knowledge Check Assignments		Oct 3 at 6:26pm		0 / 100
Module 7 Knowledge Check Assignments		Sep 29 at 2:17pm		0 / 100
Module 8 Knowledge Check Assignments		Oct 3 at 6:23pm		0 / 100

Total: N/A

Show All Scores

Show All

Course assg weighted.

Calculate graded a

You can view on What-If: know how g by upcomin; assignment; for an assign includes a s that has yet

Reflection: what I learned and how I applied it

- Cloud analytical databases (Redshift) use a separation of storage/compute patterns and prioritize analytical queries.
- Staging raw data before curated models – similar to warehouse layering.
- SQL-centric analysis on large tables and the need for clear schemas and load practices.

How I applied this to the group project

I used that idea in our group project as technical lead for the data platform. We are not using a single table to structure PostgreSQL, but rather three layers: staging is the data that is cleaned from AEMO landing data in the NEMOSIS staging zone, the DWH is the panel and engineered features that are used to model the data, and the datamart and analytics schemas are the aggregates and outputs that are used for dashboards. This separation ensures that raw extract data remains separate from curated features and from tables. I also leveraged the lab's focus on controlled loading of our dual pipelines: one for a strategic full history load, an operational daily path load which only upserts the current day's data (date + 1) so as to ensure history remains intact while new data is added cleanly. Those curated layers are then queried by the operational and strategic dashboards, similar to the lab pattern of analysing a prepared warehouse layer of data, not raw source files each time.

Student: Dipesh Sedhai Student ID: S395457 Group: Danala Group 8 (Theme 2)

Project: Australian NEM Price and Demand Analysis and Forecasting System (Docker, Airflow, PostgreSQL, Streamlit)

1. Responsibilities

- Design and migrations of database schemas for layered Postgres warehouse: (staging, database warehouse, datamart, analytics).
- Owns warehouse transformation layer: Upserting to the warehouse from data feeds, building a panel/feature, and creating a daily/monthly datamart.
- (clean_data): Inspection and cleaning of raw data; (lags): Future data look back; (rolling stats): Look back and forward at a specific time; (calendar): Generate calendar data; (AEMO error): Add errors from AEMO data.
- The content of the operational / daily dashboard, meaning, actions and explanation of UI

2. Work completed since Assessment 2

- Created warehouse pipeline (committing staging upserts, panel & features tables, daily/monthly datamart loads feat(warehouse) commit)).
- Cleared up and documented cleaning procedures: snake_case rename, INTERVENTION filter, to_numeric, dropna, dedup, sort, defaults.
- Model specific features include demand/price lags (1, 12, 288), rolling mean/std, netinterchange lag, hour/DOW/weekend and aemo_demand_error.
- Developed Operational dashboard narrative: increase in RRP, peakers, demand share, interchange stress, industry/customer actions, daily upsert value.

3. Evidence links

- GitHub: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>
- Jira : <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list?jql=project==KAN+ORDER+BY+cf%5B10019%5D+ASC&atOrigin=eyJpIjoiYmlzMTBjN2JmZTQ2NGU2N2I0MGQ1MjUzZGMzMmZmYjUiLCJwIjoiaj9>
- **Key commits:**
 - feat(db): Postgres schemas staging/dwh/datamart/analytics and migrations
 - feat(warehouse): staging upserts, panel/features build, daily/monthly datamart
- Supporting artefacts: Assessment 3 PPT (raw/clean/FE/ops slides); SETUP/README in repo

4. Planned work before Assessment 4

- Final verification of schema migrations and datamart population under full strategic + operational load paths.
- Polish operational dashboard demo narrative and screenshots for the final report.
- Support evaluation write-up linking features and cleaning choices to model behaviour.
- Contribute to final README one-command demo and individual contribution evidence pack.

Page 2 of 2 — Lab evidence and reflection

Student: Dipesh Sedhai Student ID: S395457

Lab 2 (Week 4) — Module 8: Storing and Organizing Data

Lab title: Storing and Analyzing Data by Using Amazon Redshift

13-176492 > Modules > Module 8 Storing and Organizing Data > Lab: Storing and Analyzing Data by Using Amazon Redshift

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[Version 1.0.23]

Storing and Analyzing Data by Using Amazon Redshift

Lab overview and objectives

Of the members of the data science team, Mary most often works with large datasets and columnar data stores. You have decided to explore using Amazon Redshift for your next proof of concept (POC) to address these use cases for her.

You ask Mary if she has a sample dataset that you can use for your POC. She suggests that you use a large music ticket sales dataset that she has worked with before. The dataset is available from a public Amazon Simple Storage Service (Amazon S3) bucket.

Submit Submission Report Grades

Total score 30/30

TASK 2 - Cluster created 5/5

TASK 2b - Security group associated 5/5

TASK 3 - Tables created and data loaded 5/5

TASK 4 - Table data queried 5/5

TASK 5 - Queried by using the AWS CLI 5/5

TASK 7 - IAM user queried data 5/5

Previous Next

Home	Module 2 Knowledge Check Assignments	Aug 11 at 10:33pm	0 / 100
Modules	Module 3 Knowledge Check Assignments	Aug 20 at 12:52pm	0 / 100
Discussions	Module 4 Knowledge Check Assignments	Sep 1 at 1:16pm	0 / 100
Grades	Module 5 Knowledge Check Assignments	Sep 1 at 2:29pm	0 / 100
Lucid (Whiteboard)	Module 6 Knowledge Check Assignments	Sep 29 at 11:33am	0 / 100
Notebook	Module 7 Knowledge Check Assignments	Sep 29 at 2:11pm	0 / 100
	Module 8 Knowledge Check Assignments	Sep 29 at 3:48pm	0 / 100
	Module 9 Knowledge Check Assignments	Oct 2 at 6:56pm	0 / 100
	Module 10 Knowledge Check Assignments		0 / 100
	Module 11 Knowledge Check Assignments		0 / 100

What I learned from the lab

From the lab I learned that cloud analytical databases such as Amazon Redshift are designed differently from ordinary transactional databases: they separate storage and compute patterns and favour columnar layouts that suit large analytical queries. The lab also showed why raw data should be staged before it is turned into curated models, which mirrors proper warehouse layering. Finally, it reinforced that analysis on large tables is largely SQL-centric, so clear schemas and disciplined load practices are essential if queries are to stay usable and reliable.

How I applied this to the group project

- Organised analytical storage (in contrast to stored in a single flat table as in PostgreSQL schemas staging → dwh → datamart → analytics)
- Kicked off using NEMOSIS extract as landing/staging data, cleaned and feature engineered in curatedpanel/features and datamart for ML and dashboards.
- Introduced staging upsert and daily/monthly datamart loads based on production practices of applyupsert and loading only new periods rather than the entire period.
- Both are modeled in a similar way as warehouse versus BI with operational dashboard queries using a curated table and strategic path having the same short term evidence, but a multi-year evidence using the same layeredstore.evidence.

Project: Australian NEM Price and Demand Analysis and Forecasting System (Docker, Airflow, PostgreSQL, Streamlit)

Responsibilities and work completed since Assessment 2

- **Strategic Insights dashboard (port 8502, dashboard_insights.py).** Created the Streamlit application that includes region and year filters, KPI cards (mean RRP, mean demand, rows in view, regions), and four tabs: Overview, Heatmaps, Generation and Models.
- **Automated insight cards:** Derived from the data: peak price hour (PHH), highest and lowest average regional RRP, correlation of RRP and spare capacity and highest semi-scheduled share of demand.
- **Strategic visualisations:** Produced the regional RRP time series, mean RRP by region, hours by price regime (donut chart), the regional by hour RRP heatmap with region selector, the generation and interchange comparison, and the RRP versus spare capacity scatter plot.
- **Operational Daily dashboard (port 8501, dashboard_daily.py):** Improved the dashboard by incorporating KPI cards and Plotly charts using data from the datamart, including data at daily intervals, demand share by region, average daily RRP and demand, and recent 12-hour forecasts.
- **Decision interpretation:** Provided the meaning and action guidance to the energy operators, big industry and energy customers for each strategic view to enable decision-making based on charts.
- **Presentation:** Demonstrated the Strategic dashboard section (slides 12-15) in the group demonstration.

Supporting evidence:

- GitHub repository: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>
- My branch (Dilli): <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/tree/Dilli>
- Commit 632b600, strategic insights dashboard (272 lines added): <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commit/632b600>
- Commit a92f816, operational daily dashboard: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commit/a92f816>
- Jira ticket KAN-18: <https://shrestharoshan931-1785816864381.atlassian.net/browse/KAN-18>
- Jira project board: <https://shrestharoshan9311785816864381.atlassian.net/jira/software/projects/KAN/list>

Planned work before Assessment 4

- Validate dashboard figures against the stabilized full multi-year data load.
- Add year-over-year comparisons to the Strategic dashboard to show price persistence.
- Display walk-forward evaluation results on the Models tab.
- Contribute to the final report, README documentation and testing of both dashboards.

Dilliram Dhakal5396633_Lab-2

Lab 2 (Week 4): Module 8 Storing and Organizing Data - Lab: Storing and Analyzing Data by Using Amazon Redshift. Result: 30/30, on 03-10-2026 at 15:46.

The screenshot shows the AWS Academy interface for a lab titled "Storing and Analyzing Data by Using Amazon Redshift". The lab overview and objectives are displayed, along with a task list on the right. The task list includes:

- TASK 2 - Cluster created 5/5
- TASK 2b - Security group associated 5/5
- TASK 3 - Tables created and data loaded 5/5
- TASK 4 - Table data queried 5/5
- TASK 5 - Queried by using the AWS CLI 5/5
- TASK 7 - IAM user queried data 5/5

The total score is 30/30.

The screenshot shows the AWS Academy interface for the "Grades" page. The page displays a table of grades for various assignments, including "Module 4 Knowledge Check Assignments". The table has the following columns: Name, Due, Submitted, Status, and Score. The "Module 4 Knowledge Check Assignments" row is highlighted, showing a score of 100/100.

Name	Due	Submitted	Status	Score
Course assessment Assignments				0 / 100
Module 2 Knowledge Check Assignments	Sep 2 at 3:36am			0 / 100
Module 3 Knowledge Check Assignments	Sep 2 at 4:19am			0 / 100
Module 4 Knowledge Check Assignments	Sep 13 at 11:21am			0 / 100
Module 5 Knowledge Check Assignments	Sep 29 at 2:17am			0 / 100
Module 6 Knowledge Check Assignments	Oct 3 at 6:52am			0 / 100
Module 7 Knowledge Check Assignments	Oct 3 at 6:14am			0 / 100
Module 8 Knowledge Check Assignments	Oct 3 at 6:10am			0 / 100
Module 9 Knowledge Check Assignments				0 / 100
Module 10 Knowledge Check Assignments				0 / 100

Reflection: what I learned and how I applied it

The purpose of this lab was to use Amazon Redshift, a columnar cloud data warehouse, to create a cluster, to create a security group to use with the cluster, to create tables and load data into them, and then to query data using the Amazon Redshift console, AWS CLI, and an IAM user. What I learned is that a data warehouse is intended for analytical queries over big tables and that the way data is organized and loaded influences query speed and usability. Used this knowledge to help with the group project. We have a layered approach to schemas within our PostgreSQL warehouse, with raw AEMO data stored in a staging schema, cleaned features stored in a dwh schema and aggregated tables stored in a DataMart and analytics schema, which keeps data separate from features that are cleaned and from aggregated tables that are used to drive dashboards. The two dashboards I created are based on these aggregated layers and not the raw 5-minute data, making queries simple and responsive. The lab also confirmed the benefits of incremental loading – we deliver this every day with our pipeline and delete and load one calendar day at a time.

PRT661 Assessment 3 — Individual Work and Contribution Summary

Page 1 of 2 — Group project contribution

Student: Roshan Kumar Shrestha Student ID: S395498 Group: Danala Group 8 (Theme 2)

Project: Australian NEM Price and Demand Analysis and Forecasting System (Docker, Airflow, PostgreSQL, Streamlit)

1. Responsibilities

- Strategic analytics owner for NEM project, hourly roll-ups, price regimes, and metrics for comparing models.
- Responsible for loading analysis frames into the PostgreSQL analytics schema for purposeful use.
- Support datamart / analytics load practices and datamart / analytics repository hygiene (e.g. .gitignore).
- Contribution to the analytical foundation used by the Strategic Insights dashboard (Overview, regimes, Models tab).

2. Work completed since Assessment 2

- Implemented hourly rollups from 5-minute NEM data to support multi-year strategic analysis.
- Developed price regime logic (e.g. Low / Normal / High / Spike) to classify, not just to make continuous predictions.
- Provided regression and classification metrics (such as MAE, RMSE, R^2 and regime accuracy) to compare models (Ridge / HistGB and classifiers).
- Loaded analysis frames into Postgres analytics so strategic queries and the insights dashboard can use curated tables rather than raw 5-minute extracts only.
- Earlier project support: datamart-oriented load work, branch merge, .gitignore, and Dockerfile-related updates to keep the analytics path runnable in Docker.
- Enabled the strategic narrative in Assessment 3: peak patterns, regime distribution, and honest model performance for operators and planners.

3. Evidence links

- GitHub: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>
- Jira : <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list?jql=project%3DKAN+ORDER+BY+cf%5B10019%5D+ASC&atlassian-jira-auth=eyJpOiYmlzMTBjN2JmZTQ2NGU2N2I0MGQ1MjUzZGMzMmZmYjUilCJwIjoaiJ9>

4. Planned work before Assessment 4

- In the final report enhance walk forward / evaluation write up of regression / regime models.
- Match up outputs from analytics tables with a fixed demo path (after strategic load, metrics table and regime chart of the demo is filled).
- Record the process of creating hourly rollups and regimes in order to be able to recreate results from SETUP/README.
- Make improvements to the metrics in Assessment 4 (clearly distinguish between MAE and RMSE and understand the distinction between accuracy and RMSE).

Page 2 of 2 — Lab evidence and reflection

Student: Roshan Kumar Shrestha Student ID: S395498 Group: Danala Group 8 (Theme 2)

Lab 2 (Week 4) Module 8: Storing and Organizing Data

Lab title: Storing and Analyzing Data by Using Amazon Redshift

Grades for s395498@students.cdu.edu.au [Print Grades](#)

► Start Lab ■ End Lab ⓘ AWS Details ⓘ Details ✕

Arrange By Due Date [Apply](#)

[Submit](#) [Submission Report](#) [Grades](#)

Name	Due	Submitted	Status	Score	Total score	30/30
Course assessment Assignments				0 / 100		
Module 2 Knowledge Check Assignments		Sep 13 at 8:46pm		0 / 100	TASK 2 - Cluster created	5/5
Module 3 Knowledge Check Assignments		Sep 13 at 8:52pm		0 / 100	TASK 2b - Security group associated	5/5
Module 4 Knowledge Check Assignments		Sep 13 at 9:19pm		0 / 100	TASK 3 - Tables created and data loaded	5/5
Module 5 Knowledge Check Assignments		Oct 3 at 8:29pm		0 / 100	TASK 4 - Table data queried	5/5
Module 6 Knowledge Check Assignments		Oct 3 at 8:38pm		0 / 100	TASK 5 - Queried by using the AWS CLI	5/5
Module 7 Knowledge Check Assignments		Oct 3 at 8:44pm		0 / 100	TASK 7 - IAM user queried data	5/5
Module 8 Knowledge Check Assignments		Oct 3 at 8:53pm		0 / 100		
Module 9 Knowledge Check Assignments				0 / 100		

Reflection: what I learned and how I applied it

I'm taking the concept of having organised data sources for heavy query from the Redshift style lab work I did. Heavy query after curating before data, land data first. The idea was to use that in my project with the analytics layer – that is, by rolling up 5 minutes of dispatch data into hourly grains of data, defining price regimes, and calculating regression and classification metrics in an analytics layer using PostgreSQL for strategic purposes. The same way warehouse thinkers think, the Strategic Insights dashboard and group demo is not just made of raw extracts, but it's read from pre-curated analytical outputs. The work on regimes and model metrics also highlighted the importance of maintaining the "honest" evaluation of NEM prices; an increase in price could reduce the R^2 of the model, making it less helpful for forecasting and more difficult to determine if a regime has occurred or not, than an increase in accuracy, as indicated by MAE/RMSE and regime alerts.